

KENYA
COUNTRY REPORT ON GENOME EDITING LANDSCAPE
ANALYSIS ON AGRICULTURE

MAY 2026

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LIST OF ABBREVIATIONS AND ACRONYMS

ABNE	African Biosafety Network of Expertise
ACA	Anti-Counterfeit Authority
AKU	Aga Khan University Hospital
AOCC	African Orphan Crops Consortium
ASAL	Arid and semi-arid lands
AUDA	African Union Development Agency
BMGF	Bill & Melinda Gates Foundation
CAC	Codex-Alimentarius Commission
CBD	Convention on Biological Diversity
CFT	Confined field trial
CGIAR	Consultative Group on International Agricultural Research
CIAT	International Center for Tropical Agriculture
CIMMYT	International Maize and Wheat Improvement Center
CPB	Cartagena Protocol on Biosafety
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
DNA	Deoxyribonucleic acid
DVS	Department of Veterinary Services
EAC	East African Community
EASTECO	East African Science and Technology Commission
EGU	Egerton University
EPA	Environmental Protection Agency
GALVmed	Global Alliance for Livestock & Veterinary Medicine
GDP	Gross Domestic Product
GE _d	Genome editing
GMO	Genetically modified organism
ICAR	Indian Council for Agricultural Research
ICGEB	International Centre for Genetic Engineering and Biotechnology
ICIPE	International Centre of Insect Physiology and Ecology
ICRISAT	International Crops Research Institute for the Semi-Arid Tropics
IDRC	International Development Research Center
IGI	Innovative Genomics Institute
IITA	International Institute of Tropical Agriculture
ILRI	International Livestock Research Institute
IPR	Intellectual Property Rights
ISAAA	International Service for the Acquisition of Agri-biotech Applications
JKUAT	Jomo Kenyatta University of Agriculture and Technology

KALRO	Kenya Agricultural and Livestock Research Organization
KEBS	Kenya Bureau of Standards
KECOBO	Kenya Copyright Board
KEFRI	Kenya Forestry Research Institute
KEMRI	Kenya Medical Research Institute
KEPHIS	Kenya Plant Health Inspectorate Services
KIPI	Kenya Industrial Property Institute
KU	Kenyatta University
MEFAs	Multilateral Environmental and Food Agreements
MLND	Maize Lethal Necrosis Disease
NARL	National Agriculture Research Laboratory
NBA	National Biosafety Authority
NEMA	National Environment Management Authority
NEPAD	New Partnership for Africa's Development
NRF	National Research Fund
ODK	Online Data Collection Kit
OFAB	Open Forum on Agricultural Biotechnology in Africa
PCPB	Pest Control Products Board
PCR	Polymerase Chain Reaction
PPP	Public-private partnership
PU	Pwani University
R&D	Research and development
SEKU	South Eastern Kenya University
STAK	Seed Trade Association of Kenya
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
UoN	University of Nairobi
USAID	United States Agency for International Development

EXECUTIVE SUMMARY

Background: Similar to many other African nations, Kenya struggles with food security issues due to a growing population. The situation is intensified by climate change, pests, and diseases. To combat this, Kenya is adopting biotechnology solutions, especially genetically modified technology and the latest genome editing (GE_d) methods. These technologies aim to boost yields and nutritional value by improving resilience to climate impacts and increasing resistance to pests and diseases in crops, livestock, forestry, marine, and fisheries sectors.

The African Union Development Agency–New Partnership for Africa’s Development (AUDA-NEPAD) commissioned a Landscape Analysis of Genome Editing (GE_d) for Kenya to evaluate the country’s readiness to develop, deploy, and commercialize genome-edited products. The study reviewed Kenya’s policies, regulations, institutions, infrastructure, and technical capacities related to modern biotechnology and GE_d, focusing on identifying key facilitators and obstacles across scientific, socio-economic, cultural, and governance domains. It explored opportunities where GE_d could address national priorities at scale, such as improving agricultural productivity, climate adaptation, food and nutrition security, reducing post-harvest losses, and diversifying healthy diets. Additionally, it highlighted staple and indigenous crops with high potential to enhance livelihoods. The assessment combined secondary data (published literature and institutional databases) with primary data collected via interviews, surveys, and questionnaires. Findings integrated insights on international treaties, policies, ongoing projects, capacity building, research infrastructure, priority crops and traits, intellectual property rights, and funding mechanisms.

Key findings

The Landscape Analysis revealed that Kenya is a signatory to several key international agreements, such as the Codex Alimentarius Commission (CAC), the UN Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), the Cartagena Protocol, and the Nagoya-Kuala Lumpur Supplementary Protocol on Liability and Redress, Access and Benefit Sharing. By integrating these commitments into its food and environmental management systems, Kenya supports global initiatives on food safety, climate change mitigation, biodiversity preservation, and ozone layer protection. The country also has a comprehensive regulatory framework—including policies, laws, regulations, and guidelines—that governs research, development, and the commercialization of modern biotechnology, including GE_d products.

The National Biosafety Authority (NBA), founded in 2010 under the Biosafety Act No. 2 of 2009 (revised in 2012), is responsible for regulating Biotechnology and GEd products. It works closely with other specialized regulatory agencies. In 2022, Kenya issued Guidelines to oversee GEd products. Its regulatory system is quite effective: 18 Conventional Food Trials (CFTs) and three environmental release permits for Biotechnology products, along with three environmental approvals for GEd products, have been issued. The country incorporates socio-economic factors into its regulatory approval process for biotechnology and GEd products, especially regarding environmental release and market entry. These factors include costs and market opportunities; employment and livelihoods; social impacts such as equity, consumer choice, cultural traditions, and food security; and environmental concerns such as effects on biodiversity and indigenous knowledge.

Kenya has several GEd projects, both completed and ongoing, that focus on key food crops, livestock, and traits to address major production challenges such as diseases, drought, and poor soils. Among these, crops account for 67%, followed by livestock at 25%, with no current projects on fisheries, marine, or agroforestry. The crop-focused GEd projects target traits such as Striga resistance in sorghum, control of maize lethal necrosis (MLN) to boost productivity, bacterial wilt resistance in banana, and the development of self-fertilizing maize via microbial strains that enhance nitrogen fixation in the maize root biosphere. They also include drought tolerance in maize and Arabidopsis through modulation of energy homeostasis; cyanogenic glycoside biosynthesis in cassava to reduce cyanide levels; non-rancid pearl millet seeds with a longer shelf life; reduced aflatoxin in groundnuts; pest- and disease-resistant yams; broad-spectrum bacterial blight resistance in rice; and wheat stem rust resistance. Livestock projects aim to develop vaccines for East Coast Fever (ECF) and African Swine Fever Virus (ASFV), and to create a transgenic goat as a model for genetic control of animal diseases.

Experts in GEd are distributed across five stakeholder groups in Kenya: regulatory bodies, research organizations, universities, private sector companies, and other government agencies. The current human capacity in GEd is primarily located at institutions such as the International Institute of Tropical Agriculture (IITA), the International Livestock Research Institute (ILRI), the University of Nairobi (UoN), Kenyatta University (KU), South Eastern Kenya University (SEKU), Biosciences Eastern and Central Africa (BecA), Kenya Agricultural and Livestock Research Organization (KALRO), Pwani University (PU), Jomo Kenyatta University of Agriculture and Technology (JKUAT), and Egerton University (EGU).

Multiple R&D and academic institutions in Kenya are involved in various GEd activities, including KALRO, KU, UoN, EGU, JKUAT, PU, SEKU, IITA, ILRI, GALVmed, AGTECH

Consulting Ltd, and ISAAA. The country has advanced its GEd capacity through targeted investments in training, infrastructure, and partnerships. IITA's Nairobi Campus features a cutting-edge GEd platform that trains African scientists, including Kenyans, in technologies such as CRISPR-Cas9, in partnership with IGI, AOCC, and UC Davis. Additional efforts by AUDA-NEPAD/ABNE, OFAB, and the Alliance for Science have enhanced stakeholder capacity for GEd communication and public engagement. Many universities, research institutions, and CGIAR centers have basic laboratory and greenhouse infrastructure to support GEd R&D, but funding shortages and limited access to specialized equipment persist.

The assessment identified several staple, indigenous, and commercial crops with significant potential for enhancement via GEd, focusing on traits related to productivity, climate resilience, post-harvest loss reduction, and food and nutrition security. Kenya has established institutions to manage intellectual property rights, including the Kenya Industrial Property Institute (KIPI), the Kenya Plant Health Inspectorate Service (KEPHIS), the Kenya Copyright Board (KECOBO), and the Anti-Counterfeit Authority (ACA). These bodies oversee patents, plant breeders' rights, and protections for biotechnology products. Although private-sector involvement in GEd remains limited, existing companies contribute significantly to research, product development, and commercialization through public private partnerships, thereby fostering innovation, technology transfer, and scaling efforts.

Kenya's GEd research funding is mainly externally driven and dependent on donors, primarily from the Gates Foundation, former USAID, and CGIAR/ILRI-linked programs, with additional industry investments from Corteva and Pivot Bio. USAID, which supported the Striga Smart Sorghum for Africa project, has undergone major restructuring, including large-scale furloughs and workforce reductions (RIF), with limited functions integrated with the State Department. Funding for USAID's GEd projects has been discontinued, creating significant funding gaps. To maintain sustainability, increased national funding through the National Research Fund (NRF Kenya) and other government initiatives will be essential, along with the expansion of PPP models that connect global technology developers with Kenyan research institutions and stakeholders. So far, NRF has not allocated specific funding for GEd research.

Policy implications

Short term (next 12–24 months): Include managing the ongoing raging debate and legal challenges surrounding GMOs. This is well explained by Kenya's recent lifting of the ban on GMO imports and cultivation, followed by its subsequent halting through pending court cases.

Medium to long term (2–5 years): Include leveraging biotech for national development, and trade in biotech products while ensuring compliance with international standards. Fostering sustainable agricultural practices, improved food security, and promoting economic growth through advances in biotech, especially GEd techniques.

Conclusion

Kenya's regulatory framework effectively encompasses the necessary laws, regulations, and guidelines to oversee and commercialize GMO and GEd products. As a result, the country has approved numerous GMO and GEd products for environmental release and research. Kenya has also made significant strides in enhancing GEd technical expertise through strategic investments in training, infrastructure, and partnerships. Nonetheless, the GEd R&D funding landscape remains largely dependent on external investors, resulting in a substantial funding gap and limited access to specialized equipment. Boosting funding for biotech and GEd R&D via NRF-Kenya can help close these gaps and promote sustainability.

1. INTRODUCTION

1.1. Description of Kenya's Agricultural Landscape

Agriculture is a key pillar of Kenya's economy. Including forestry and fisheries, it makes up about 20–25% of the gross domestic product (GDP) and roughly 18% of employment. Major agricultural exports such as tea, coffee, cut flowers, fruits (especially avocados), and vegetables account for nearly half of the country's foreign exchange earnings. Despite its importance, the sector faces challenges such as unpredictable weather, land degradation, and climate variability. To address these issues, it is necessary to adopt sustainable farming practices and climate-smart technologies that enhance productivity, resilience, and environmental health. Thus, biotechnology, especially gene editing (GE_d), is being explored as a promising solution, offering ways to improve yields in crops, livestock, forestry, marine, and fisheries. Benefits also include better nutritional quality, greater climate resilience, and enhanced pest and disease resistance.

1.2. National Frameworks and Investments

For biotechnology overall, especially genome editing, achieving significant impact in Kenya depends on strong national frameworks that promote responsible development and sustainable use. Kenya has put in place various regulatory structures, including biotechnology policies, biosafety laws, and GE_d guidelines. These are meant to balance the advantages of biotechnology with ethical, environmental, and biosafety concerns. Additionally, they create a favourable legal and administrative environment that encourages local and international investments in biotechnology and GE_d R&D. Although progress has been made, there remains substantial room to increase investment in agricultural biotechnology and GE_d infrastructure to boost productivity, innovation, and resilience in the sector.

1.3. Engagement in Regional Frameworks

Kenya is actively involved in creating and implementing regional biotechnology policies, especially within the East African Community (EAC). The country leads in developing a 10-year unified biotechnology strategy for the region. Sponsored by the East African Science and Technology Commission (EASTECO), this strategy will steer biotechnology development from 2026 to 2036. Its main aim is to standardize policies and regulations across East Africa, reducing fragmented national approaches that obstruct innovation and complicate cross-border collaboration and trade.

1.4. Challenges and Opportunities: Justification of GEd Assessment based on Current Issues Identified in Kenya

1.4.1. Challenges Justifying GEd Assessment

Despite the significant potential of biotechnology and genome editing, several challenges hinder their widespread adoption in Kenya. These include resource limitations, gaps in technical expertise and public perception issues.

Key challenges include:

- **Resource and expertise gaps:** The implementation of advanced genome editing technologies requires substantial resources and specialized human capacity, both of which remain limited in Kenya.
- **Regulatory frameworks:** Although Kenya has the Biosafety Act, Biosafety Regulations, and GEd Guidelines, challenges remain in ensuring consistent implementation and full harmonization with international standards to support biotechnology and genome editing R&D and commercialization.
- **Public perception and awareness:** Public awareness and effective communication regarding genome editing technologies remain limited, affecting acceptance and informed decision-making.
- **Technological and infrastructure limitations:** Insufficient infrastructure and limited sustained investment from both public and private sectors constrain biotechnology and GEd R&D. Increased stakeholder participation and investment are required to strengthen national ownership of genome-edited products and ensure access to enabling technologies such as CRISPR-Cas systems and laboratory materials.
- **Biodiversity conservation:** Kenya faces the challenge of balancing the application of biotechnology and genome editing with the protection of its rich biodiversity. While safeguarding ecosystems is essential, maintaining this balance in policy and decision-making remains a key challenge to the broader adoption of these technologies.

1.4.2. Opportunities Supporting GEd Investment

Biotechnology has the potential to deliver benefits aligned with Africa's needs, rather than being driven primarily by external interests. In Kenya, biotechnology and GEd offer investment opportunities, particularly in developing and commercializing climate-resilient, pest- and disease-resistant, and nutritionally enhanced crops and livestock, as well as improving tolerance to abiotic stresses such as drought and low soil fertility (Wambugu & Kamanga, 2014).

The key challenges that GEd can address include:

- **Improved food security:** Genome editing can accelerate the development of crop varieties and livestock breeds that are resistant to diseases, pests, and tolerant to abiotic stresses such as drought and heat, which are critical for enhancing Kenya's food security and agricultural resilience.
- **Enhanced nutritional value:** Through biofortification, GEd can increase the vitamin, protein, and mineral content of staple crops, thereby contributing to the reduction of malnutrition.
- **Increased crop and livestock productivity:** GEd technologies, such as CRISPR-Cas9, can enhance crop yields by improving traits such as tillering, branching, and grain number in crops including rice, sorghum, wheat, and sugarcane. In livestock, GEd technologies can improve productivity, for example, by increasing egg production in poultry and improving both the quality and quantity of milk in dairy animals.
- **Faster and more efficient breeding:** Genome editing enables precise, targeted modifications of existing genes, significantly reducing the time and cost associated with conventional breeding methods.
- **Targeted pest and disease resistance:** Ongoing research includes developing genome-edited bananas resistant to bacterial wilt and sorghum resistant to the parasitic weed Striga. In livestock, genome-editing research in Kenya is also contributing to the development of vaccines against diseases such as East Coast Fever and African Swine Fever.
- **Climate adaptation:** Genome editing provides precise tools for developing crops and livestock with enhanced tolerance to environmental stresses associated with climate change.
- **Development of Centers of Excellence: GEd and modern biotechnology in general provide an opportunity to invest in** establishing research hubs in universities and national agricultural research institutions (NARIs) to foster innovation and talent.

1.5. Objectives of the Landscape Analysis

The overall objective of the Genome Editing Landscape Analysis was to conduct an in-depth assessment of existing policy, institutional, infrastructural, and technical capacities that support product development and commercialization in selected African countries.

Specifically, for Kenya, the objectives of the Landscape Analysis were to:

- i. Provide an evidence-based assessment of the status of modern biotechnology and genome editing, highlighting key trends, enabling and constraining factors, and priority

areas for attention. This includes scientific and technical aspects, as well as political, geopolitical, social, cultural, and human dimensions that influence the application of genome editing in agriculture and food systems.

- ii. Identify emerging national needs that genome editing can effectively address, particularly those requiring rapid and scalable responses within food systems, including agricultural productivity, reduction of post-harvest losses, climate adaptation, food and nutrition security, and diversified and healthy diets.
- iii. Identify priority staple and indigenous crops within Kenya's national context that have the potential to improve livelihoods through enhanced food security, nutrition, climate resilience, and sustainable productivity. These include crops such as maize, wheat, rice, potato, beans, and garden peas.

2. APPROACH/METHODOLOGY

Data were collected from both secondary and primary sources. Secondary data were gathered from published literature and institutional databases, including official reports, peer-reviewed publications, and relevant organizational websites. Primary data were collected through live interviews using the Online Data Collection Kit (ODK), structured surveys, and email-based questionnaires shared in Microsoft Word format. Primary data collection served to validate, supplement, and strengthen the secondary data. Data from both sources were subsequently analysed, synthesized, and packaged under the following thematic areas:

2.1. Regulatory and Policy Frameworks for Biotechnology and Genome Editing

Components of the biotechnology and genome-editing regulatory and policy frameworks were identified and documented using secondary sources (published literature and institutional websites) and primary data from interviews, surveys, and email correspondence.

2.2. Genome Editing Projects

Information on GEd projects was compiled from primary and secondary data sources. This included documentation of target crops, livestock, fisheries, and forestry species; traits under development; key stakeholders and partnerships involved; and funding sources. The synthesized data were used to:

- i. identify emerging economic, social, and environmental/climate-related needs; and
- ii. assess the status of human capacity and infrastructure supporting GEd technologies in Kenya.

2.3. Traditional and Staple Crops for Genome Editing Improvement

Data on genome editing projects were further disaggregated to identify traditional and staple crops with the highest potential for socio-economic impact. Selection criteria included national acceptance, resource requirements, scalability, and likelihood of successful development and deployment.

2.4. Institutional Capacity (Human Capital, Infrastructure, and Equipment)

Data on institutional capacity, including human resources, laboratory and field infrastructure, and equipment for GEd R&D, commercialization, and scaling, were collected from respondents during the primary data collection exercise.

2.5. Stakeholder Mapping

Key stakeholders and institutions, and, where appropriate, individual experts, were identified to provide critical information on existing biotechnology and GEd interventions. Stakeholders were categorized into five (5) groups, as reflected in the data collection tools:

- Regulatory agencies
- Research organizations and institutions
- Universities
- Private sector and industry
- Government ministries, agencies, and policymakers

NB. List of key stakeholders and their institutions in Kenya are annexed (Annex 1).

2.6. Database Systems and Data Management

Customized data collection tools (questionnaires) and digital platforms were developed to support primary data collection. Questionnaires were tailored to each stakeholder category and used to generate structured datasets assessing Kenya's preparedness and capacity to adopt, implement, and scale GEd technologies.

Enumerators were identified, recruited, and trained through an online induction program on the use of ODK and associated data-collection tools. Training was conducted by the IT team and train-the-trainers (ToT) from the Africa Harvest and AGTECH Consortium. All tools and platforms were pre-tested prior to deployment.

Both methodological and data triangulation approaches were employed to reduce bias and enhance the validity and reliability of findings. All collected data were encrypted and stored on a secure, centralized server to ensure participant confidentiality and compliance with ethical standards.

2.7. Data synthesis and Data Analysis

Data collected was synthesized and packaged into appropriate information, and, where applicable, tables were generated.

Variables investigated included, but were not limited to:

- i. signatory to multinational food and environmental agreements;
- ii. regulatory framework components;
- iii. applications received and approved (field testing, registration, and commercialization);
- iv. R&D institutions, personnel, and laboratory/field facilities;
- v. projects;
- vi. crops and livestock species; and
- vii. traits under GE development.

2.8. Interactive Map

An interactive map, comparable to the Agenda 2063 Dashboard, was developed to visualize genome editing activities and capacities across Africa and is accessible at:

<https://genome.africa/>

3. RESULTS

3.1. Membership in Multilateral Environment Treaties and Agreements

Kenya has signed numerous international treaties and agreements related to the environment and food safety, such as the Codex Alimentarius Commission (CAC), the UN Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), the Cartagena Protocol on Biosafety, and the Nagoya–Kuala Lumpur Protocol on Liability, Redress, Access, and Benefit Sharing (see *Table 1*). By ratifying these treaties and incorporating them into its national policies, Kenya commits to global initiatives on food safety, climate change, biodiversity, and environmental protection. Kenya has been a member of the CAC since 1969. The Kenya Bureau of Standards (KEBS) acts as the national authority and Codex Contact Point, overseeing the alignment of Kenya's food safety standards with international Codex standards to ensure food safety, promote fair trade, enhance access to global markets for Kenyan exports, and safeguard consumer health.

Kenya signed the UNFCCC in 1992 and ratified it in 1994. Similarly, it signed the CBD on June 5, 1992, at the Rio Earth Summit, and ratified it on October 24, 1994. In line with the CBD, Kenya also developed its National Biodiversity Strategy and Action Plan (NBSAP) that same year. Regarding the Cartagena Protocol on Biosafety, Kenya was the first country to sign the Protocol on May 15, 2000, and ratified it officially on September 11, 2003. The National Biosafety Authority (NBA) acts as the national competent authority for the Protocol. Kenya also became a Party to the Nagoya–Kuala Lumpur Supplementary Protocol on Liability and Redress, signing it in 2010, acceding in 2024, and bringing it into force in 2025.

Table 1: Status of Kenya's participation in key multilateral environmental and food agreements (ME&FAs) or treaties

Multilateral Environmental & Food Agreements (MEFAs) / Treaties	Date: Signatory & Ratification or Accession by the Country	Reference
Codex Alimentarius Commission (CAC): UN Food and Agriculture Organization and World Health Organization (UN FAO-WHO), a joint established body to develop international food standards, guidelines, and codes of practice <i>critical for risk assessment of food developed through GEd</i>	1969 Member	https://www.fao.org/fao-who-codexalimentarius/about-codex/members/en/

UNFCCC convention on climate change	Signed 1992 Ratified 1994	https://unfccc.int/resource/docs/natc/kennc2es.pdf
Convention on Biological Diversity (CBD)	1992 Signed 1993 Ratified	https://www.cbd.int/doc/legal/cbd-en.pdf
Cartagena Protocol on Biosafety (CPB)	2000 Signed 2003 Ratified	https://www.biosafetykenya.go.ke/images/COPMOB-Report.pdf
Nagoya Protocol on Liability & Redress, Access & Benefit Sharing (ABS)	Signed 2010 Ratified 2014	https://www.cbd.int/abs/nagoya-protocol/signatories

3.2. Policies that Advance or Hinder Adoption of Modern Biotech and GE

Kenya has two policies that promote modern agricultural technology, including GE. These policies are: the National Biotechnology Development Policy (2006) and the Agricultural Policy 2021.

- **National Biotechnology Development Policy (2006)** - lays the groundwork for the application of modern biotechnology, including GE, to improve crops (pest resistance, drought tolerance, etc.) and livestock (vaccines, productivity, etc.). It established the need for a safe, structured, and legal system to handle GMOs and directly gave rise to the country's regulatory framework.
- **Agricultural Policy 2021** promotes the use of modern technology, including biotechnology, to boost agricultural productivity while emphasizing sustainable resource use. This policy aligns technology, including modern biotechnology, with Vision 2030.

No policy was found in Kenya during this Assignment that hinders or may hinder the adoption of modern biotechnology and GE.

3.3. National Regulatory Framework

3.3.1. Biosafety Act/Law, Biosafety Regulations/Decrees and GE Guidelines

Kenya's regulatory framework for the governance of biotechnology and genome-edited products comprises the Biosafety Act No. 2 of 2009 (Revised Edition 2012), the Biosafety (Contained Use) Regulations, 2011, the Biosafety (Environmental Release) Regulations, 2011, the Biosafety (Import, Export and Transit) Regulations, and the Guidelines for Determining the Regulatory Process of Genome-Edited Organisms and Products in Kenya (2022) (See Table 2). Overall, Kenya's regulatory framework is considered functional and enabling.

The National Biosafety Authority (NBA) is the principal competent authority responsible for regulating biotechnology and genome-edited products, working in close consultation with

other specialized regulatory agencies. For genome editing, four key specialized regulatory agencies were identified in this assessment from a broader list of eight. These include:

- Kenya Plant Health Inspectorate Service (KEPHIS) – plant products and growth stimulants;
- Department of Veterinary Services (DVS) – livestock and veterinary products;
- Kenya Bureau of Standards (KEBS) – conformity of products to national standards; and
- Kenya Industrial Property Institute (KIPI) – protection of intellectual property rights.

The NBA, in consultation with specialized regulatory agencies, would receive applications, check for completeness, conduct risk assessments, decide whether to approve them, and communicate its decision to the applicant. The regulatory review processes of biotech products are presented in Figure 1 (contained use), Figure 2 (field trials), and Figure 3 (environmental release).

Application, Review & Approval (Contained use)

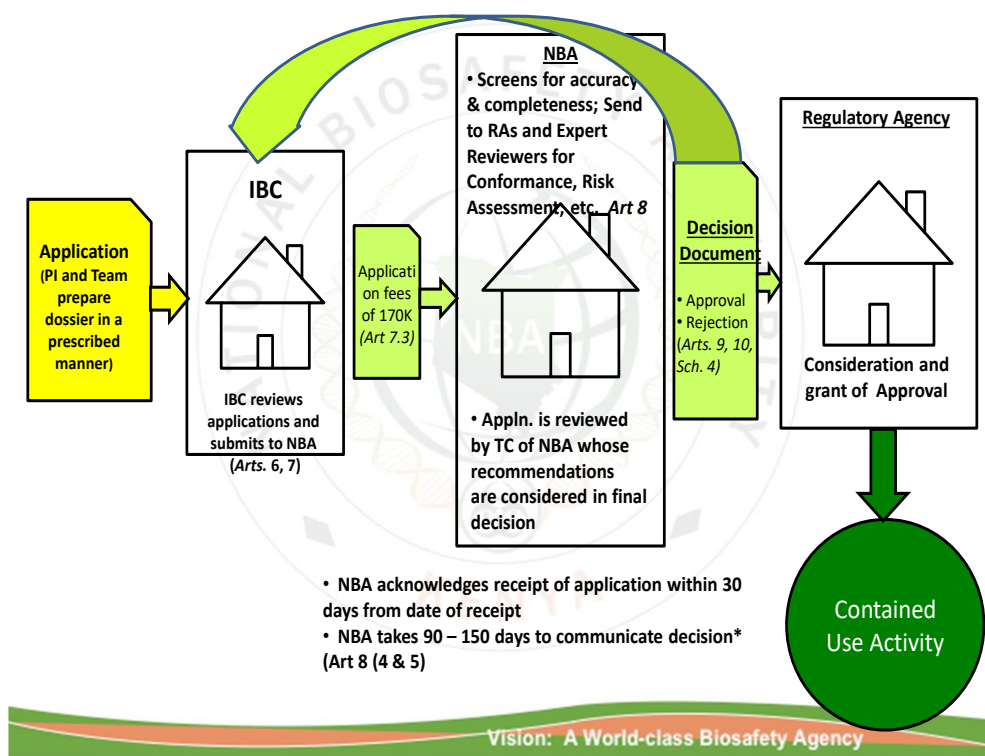


Figure 1: Application review process for contained use

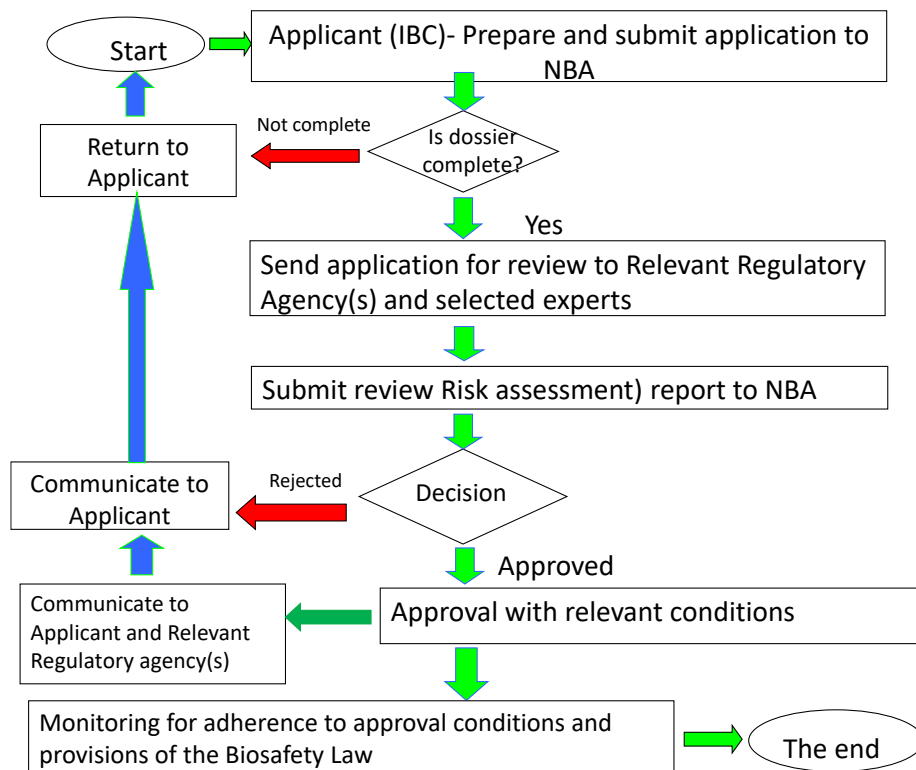


Figure 2: Application review process for field trials

Table 2: Regulatory and institutional landscape for genome editing in Kenya

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference
National Biosafety Authority (NBA)	Transfer, handling, and use of GEd and Biotech Products	A National Biotechnology Development Policy 2006	2006	R&D, Commercialization, Trade, etc.	www.biosafetykenya.go.ke
		Biosafety Act No.2 of 2009	2009	Contained use, environmental release, import, export, transit, and placing on the market of GMOs	www.biosafetykenya.go.ke
		Biosafety Contained Use Regulation 2011	2011	Contained use of GMOs	www.biosafetykenya.go.ke
		Biosafety Environmental Release Regulations 2011	2011	Environmental release and placing in the market of GMOs	www.biosafetykenya.go.ke
		Biosafety Import, Export, and Transit Regulations 2011	2011	Import, export, and transit of GMOs	www.biosafetykenya.go.ke
		Biosafety Labelling Regulations 2012	2012	Labelling of GMOs on the shelves in the market	www.biosafetykenya.go.ke
		Guidelines for determining the regulatory process of genome-edited organisms and products in Kenya	2022	All genome-edited organisms (GEs or GEOs) and derived products	www.biosafetykenya.go.ke
Kenya Plant Health Inspectorate Service (KEPHIS)	Plant varieties, seed quality & health	Kenya Plant Health Inspectorate Service Act	2012	Plant Health and Quality Assurance	www.kephis.go.ke
		Plant Protection (General) Regulations	2021	Prevention of pest introductions and spread, management of pests and weeds,	www.kephis.go.ke

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference
				surveillance of plants and plant products, and regulation of plant movement and trade	
		Plant Protection Act Cap 324	2022	Regulate the movement of plants and related items; empower KEPHIS to control the cultivation and harvesting of specific plants in affected areas; enforce measures to prevent the spread of pests; and license and inspect nurseries.	www.kephis.go.ke
		Seeds and Plant Varieties Act Cap 326	2022	Testing, certification, and regulation of seeds and plant varieties to ensure quality, prevent the sale of fake seeds, and establish a national index of approved varieties	www.kephis.go.ke
		Plant Breeders Rights and Regulations	2022	Grant the right to plant breeders, upholding variety protection, enforce standards to ensure quality seeds reach farmers, and prevent the introduction of pests and diseases	www.kephis.go.ke
Department of Veterinary Services (DVS)	Animal diseases and pests	Animal Diseases Act Cap 364, 1965 (revised 2007, 2012, 2022)	1965, 2007, 2012, 2022	Prevent, control, and eradicate animal diseases, with specific provisions for the Directorate of Veterinary Services (DVS) to declare infected areas, impose movement restrictions, and regulate the importation of animals and animal products	www.kilimo.go.ke

Institutions	Mandate / Relevance to GEd	Regulatory instruments	Date of enactment or publication	Coverage/ scope	Reference
Kenya Industrial Property Institute (KIPI)	Intellectual Property Rights protection	The Industrial Property Act Cap 509, 2001 (revised 2002, 2007, 2017, 2018, 2022)	2001	Promoting innovation and facilitating the acquisition of technology and providing a legal framework for the protection and regulation of industrial property rights, including patents, utility models, and industrial designs.	www.kipi.go.ke
		Industrial Property Regulations Guidelines for the examination of Patents, Utility Models and Industrial Designs	2000	Trade liberalization, dismantling of import licensing, adoption of pre-shipment inspection, and a managed floating exchange rate system in the year 2000.	www.kipi.go.ke
		Industrial Property Rights Guidelines for Commercialization in Kenya	2007	Provide guidance on procedures for examining and granting patents, utility models, and industrial designs, and ensure consistent application of the law.	www.kipi.go.ke
Kenya Bureau of Standards (KEBS)	Standards (commodities, materials & substances)	Standard Act	Rev. 2012	Promote standardisation, providing conformity assessment services (e.g. testing, certification), maintain International Systems of Unit (SI), etc.	www.kebs.org

3.4. Socio-Economic Considerations for Decision-Making

Kenya has integrated socio-economic considerations into regulatory decision-making processes for biotechnology and genome-edited products, particularly during environmental release approvals and market placement. Such considerations are incorporated through public participation mechanisms and play a critical role in determining whether biotechnology and GEEd products are advanced or constrained in adoption. These considerations encompass economic, social, cultural, human, and environmental dimensions, including costs and market opportunities, employment and livelihoods, equity, consumer choice, food security, and impacts on biodiversity and indigenous knowledge systems as elaborated below:

Economic impacts include:

Employment and livelihoods: Adoption of biotech/GEEd products may create new employment opportunities, though some stakeholders' express concerns about potential job displacement.

Market opportunities: Biotechnology and GEEd can open new markets, while critics raise concerns about market substitution and impacts on traditional products.

Costs of inputs: The affordability of biotech/GEEd-derived inputs, such as seeds, is a key concern, particularly for resource-poor farmers.

Social and cultural impacts include:

Equity: Ensuring fair distribution of benefits and costs across different population groups, including gender and age considerations.

Consumer choice: Access to information and the ability of consumers to make informed choices.

Cultural practices: Potential impacts on traditional seed systems, indigenous knowledge, and local customs that require careful evaluation to avoid the erosion of cultural heritage.

Food security: Assessment of how biotech/GEEd products influence food availability, access, and stability.

Environmental considerations include:

Impacts on biodiversity and traditional crops: Potential displacement of traditional varieties and loss of genetic diversity, which may have direct socio-economic implications.

Impacts on indigenous technologies: Concerns that the replacement of traditional farming methods could disrupt local agricultural systems and associated cultural practices.

3.5. Factors that Hinder or Advance Application of Modern Biotech and GEd

Political Factors: Like in many countries in Africa, the adoption of modern biotechnology in Kenya is dictated by political will. For example, in 2012, the National Rainbow Coalition (NARC) government banned the cultivation and use of GMOs, whereas in 2022, the Kenya Kwanza government (elected in 2022) lifted the 10-year ban, allowing the country to embrace the use of GMOs to combat climate change (drought) and food insecurity.

Geopolitical Factors: Most of Kenya's biotech R&D and capacity-building is funded by foreign/external investors (the Gates Foundation, former USAID, European Union Member States, etc.). This creates a situation where modern biotechnology, including GEd, is heavily influenced by foreign entities, resulting in a donor-driven approach rather than a purely national one. In certain cases, especially with EU member states, the donor pushes for the Precautionary Principle, which could lead to revising existing, evidence-based regulations/guidelines, making them more restrictive or prohibitive.

Social and Human Factors: A significant portion of the Kenyan public fears that modern biotechnology (GMO) products are not safe, particularly concerning toxicity or allergies. Critics of the technology also express widespread concern that its products will harm biodiversity by damaging ecosystems, including beneficial insects and indigenous crop varieties. On seed sovereignty, critics argue that the technology will force smallholder farmers to buy expensive, proprietary seeds from multinational corporations every season rather than save their traditional or landrace varieties.

Culture and Traditions: Certain cultural beliefs and traditions regarding "natural/organic" foods conflict with the concept and application of modern biotechnology in Kenya.

Sociocultural, human, public health and food sovereignty concerns have led to a litany of court cases and legal challenges since the lifting of the 10-year GMO ban in 2022. These court cases and legal challenges include:

- **High Court of Kenya at Nairobi (2022/2023):** Several petitions filed by stakeholders, including the Kenyan Peasants League (KPL) and other civil society actors, secured initial conservatory orders prohibiting the government from importing or distributing GMOs.
- **Law Society of Kenya (LSK) v. Attorney General (2023/2024):** The LSK, along with other groups, challenged the Cabinet's decision to lift the ban, arguing that it was made without sufficient public participation and violated consumer rights. This petition was initially dismissed in 2023 by the Environment and Land Court (ELC), which cited a robust regulatory framework. It was held that the NBA was complying with the required regulations. However, this ruling was later subjected to further appeals and caught up in the 2025 appellate review.

- *Kenya Peasants League (KPL) v. Attorney General & 4 Others (2025)*: On March 7, 2025, the Court of Appeal issued a conservatory order halting the importation and cultivation of GM crops in Kenya, reversing a high court victory for the government. The appellant argued against the move on constitutional, human health, and environmental grounds.
- **Seed Savers Network and Others Case (2025)**: On November 27, 2025, the High Court ruled in Favor of 15 smallholder farmers and other organizations, including Seed Savers Network, the Biodiversity and Biosafety Association of Kenya (BIBA), and Greenpeace Africa. The court declared that parts of the Seeds and Plant Varieties Act are unconstitutional for criminalizing the exchange of indigenous seeds.

3.6. Genome Editing Programmes and Projects

Several GEEd projects, both completed and ongoing, have been identified in Kenya, focusing on key food crops, livestock, and economically important traits, such as disease resistance, drought tolerance, and adaptation to poor soils (Table 3).

Crops account for the majority of genome editing projects (67%), followed by livestock (25%). Currently, no GEEd initiatives have been identified in fisheries, marine systems, or agroforestry. In terms of project maturity, most initiatives are at the research and development stage, with only one product, the East Coast Fever vaccine, having progressed to the scaling phase.

Table 3: Genome editing projects and programmes in Kenya

Projects/ Programs (organism)	Trait	Collaborating partners	GE ^d Technique	Stage (Lab, field trial, commercialization)	Funding (US\$)	Funding source	Reference/lead scientist
Striga Smart Sorghum for Africa Project	Striga resistance	Kenyatta University (Lead), Addis Ababa University (Ethiopia), International Service for the Acquisition of Agri-biotech Applications (ISAAA)	CRISPR/Cas9	Research & Development (R&D)	3.8	Feed the Future of the United States Agency for International Development (USAID)	Ngure & Karembu, 2023
Gene editing to control maize lethal necrosis (MLN) disease in Africa for improved maize productivity and grain harvests	MLN	CIMMYT (Lead), Corteva Agriscience, Kenya Agricultural & Livestock Research Organization (KALRO) and Seed Trade Association of Kenya (STAK)	CRISPR-Cas9	Deployment & Commercialization (D&C)	Not provided	Bill & Melinda Gates Foundation (BMGF)	Ngure & Karembu, 2023
Genetic improvement of banana for control of bacterial wilt disease	Bacterial wilt resistance	International Institute of Tropical Agriculture (IITA), Nairobi, Kenya (Lead), National Agriculture Research Laboratory (NARO-NARL), Kawanda,	CRISPR/Cas9	R&D	Not provided	USAID and the Roots, Tubers and Bananas program of CGIAR	Tripathi <i>et al.</i> , 2022

Projects/ Programs (organism)	Trait	Collaborating partners	GE ^d Technique	Stage (Lab, field trial, commercialization)	Funding (US\$)	Funding source	Reference/lead scientist
		Uganda					
PROVEN [®] 40 Microbial Field Efficacy Trials in Kenya	Self-fertilizing (Nitrogen)	AGTECH Consulting Ltd (AGTECH), Kenya Agricultural & Livestock Research Organization (KALRO) and Pivot Bio, Inc., USA	Suicide plasmid	D&C	Not provided	Pivot Bio, Inc.	Micheni <i>et al.</i> , 2023* (*Technical Report)
Modulation of energy homeostasis in maize and Arabidopsis to develop lines tolerant to drought, genotoxic, and oxidative stresses	Drought tolerance	Kenyatta University, Plant Transformation Laboratory (Lead), Kenya, VIB-University of Gent, Center for Plant Systems Biology, Belgium	CRISPR/Cas9	R&D	Not provided	Belgian Development Corporation	Njuguna <i>et al.</i> , 2018
Role of the CYP79D1 gene in biosynthesis of cyanogenic glycosides in cassava using the CRISPR/Cas9 genome editing system	Low cyanogenic glycosides biosynthesis	Kenyatta University (Lead), Jomo Kenyatta University of Agriculture and Technology, Pwani University, South Eastern Kenya University	CRISPR/Cas9	Discovery & Innovation (D&I)	Not provided	The World Academy of Sciences and the International Centre for Genetic Engineering & Biotechnology	Juma <i>et al.</i> , 2022

Projects/ Programs (organism)	Trait	Collaborating partners	GE^d Technique	Stage (Lab, field trial, commercialization)	Funding (US\$)	Funding source	Reference/lead scientist
Introducing the non-rancidity trait into pearl millet seed / Increasing shelf life of pearl millet milled flour	Non-rancidity	ICRISAT, CIMMYT	CRISPR/Cas9	Discovery & Innovation (D&I)	Not provided	Bill & Melinda Gates Foundation (BMGF)	Muindi <i>et al.</i> , 2023
Gene Editing for Reducing Aflatoxin in Groundnuts	Aflatoxin Resistance	CIMMYT, ICRISAT	CRISPR/Cas9	Discovery & Innovation (D&I)	Not provided	Bill & Melinda Gates Foundation (BMGF)	Prasad <i>et al.</i> , 2023
CRISPR/Cas-based genome editing for yam improvement	Genetic improvement for pest, disease, and yield	IITA, University of Nairobi, Iowa State University, Ames, IA, USA, Division of Plant Sciences, Bond Life Sciences Center, University of Missouri, Columbia, MO, USA, Donald Danforth Plant Science Center, St. Louis, MO, USA	CRISPR/Cas9	Discovery & Innovation (D&I)	Not provided	US National Science Foundation	Syombua <i>et al.</i> , 2021
CRISPR/Cas9 gene editing of Theileria parva for the	ECF vaccine	ILRI Nairobi (Lead), FLI (Germany), J. Craig	CRISPR-Cas9	Scaling	Not provided	CGIAR Trust Fund, USAID, BMGF	Ngure & Karembu, 2023

Projects/ Programs (organism)	Trait	Collaborating partners	GE^d Technique	Stage (Lab, field trial, commercialization)	Funding (US\$)	Funding source	Reference/lead scientist
development of a vaccine against East Coast fever (ECF)		Venter Institute (JCVI), USA					
Accelerating African Swine Fever Virus (ASFV) vaccine development via CRISPR-Cas9 and synthetic biology technologies	ASFV vaccine	ILRI Nairobi (Lead), FLI (Germany), J. Craig Venter Institute (JCVI), USA	CRISPR-Cas9	R & D	Not provided	CGIAR Trust Fund, USAID, BMGF	Ngure & Karembu, 2023
Application of reproductive biotechnologies to develop a transgenic goat as a model for genetic control of animal diseases	Trypanosomiasis disease	ILRI (Lead), UoN	CRISPR-Cas9	D&I	Not provided	CGIAR Trust Fund	Deeni & State, 2024
Introduction of broad-spectrum bacterial blight resistance in rice	Bacteria blight	KALRO/Japan	CRISPR-Cas9	R&D	Not provided	BMGF	Emily Gichuki (January 2024). (personal communication)

3.7. Human Capital and Institutional Capacity

3.7.1. Summary of Existing Human Capacity

Kenya possesses a growing pool of human capacity in GEd regulation and R&D, particularly in agricultural systems. Expertise was identified across five main stakeholder categories: regulatory agencies, research institutions, universities, private sector entities, and government ministries and policymakers (See *Annex 1*). Notable capacity-building initiatives include curriculum integration at Kenyatta University and collaborative GEd platforms established by the IITA and the IGI at the ILRI. These initiatives train African scientists in cutting-edge genome editing technologies.

Key experts in genome editing in Kenya include Leena Tripathi and Rose Harriet Okech from IITA, based at ILRI in Nairobi; Stephen Runo from Kenyatta University (KU); and Christian Tiambo and Kouadio Nasser Yao from Biosciences Eastern and Central Africa (BecA-ILRI). Other scientists who possess extensive knowledge in GEd include Martin Kiogora Mwirigi, James Kamau Karanja, Emily Waringa Gichuhi, Paul Kuria, and Jane Mbugua from KALRO; Edith Mnyazi Muwawa and Prof. Muema from Pwani University; Grace Wacheke Mungai from Jomo JKUAT; and Stephen Kironji Githeng'u from Egerton University.

Despite this strong base, capacity gaps remain, including limited investment in advanced laboratory infrastructure (e.g., high-throughput sequencing), insufficient numbers of trained scientists and regulators, the need for additional hands-on training platforms, and inadequate funding to sustain GEd.

3.7.1.1. Women, Youth and Special Interest Groups' Involvement in Modern Biotech and GEd

This group is underrepresented in Kenya's biotech R&D, which remains male-dominated, with women's involvement estimated at 18-33%. However, the gap is being narrowed through specialized training programs such as the IITA platform at ILRI Nairobi, targeted funding, and infrastructure upgrades, fostering women's and youth-led innovations in the agriculture and health sectors. For example, the sample data gathered on key experts in GEd in this Assignment indicates that 6 out of 14 (approximately 40%) GEd research experts are women, whereas 60% are men. This data was not disaggregated further to ascertain the fraction that is due to youth and special interest groups, like persons with disabilities. However, in Kenya, youth and persons with disabilities are increasingly being encouraged to participate in tech-driven solutions, including modern biotechnology applications, through the Kenya Research

Financing and Capacity Strengthening Masterplan (2026 - 2036) and the Momentous Pilot Fund, launched in March 2026 by the Assistive Technologies for Disability Trust (AT4D).

3.7.2. Research, Development and Academic Institutions

Several R&D and academic institutions in Kenya are engaged in various activities of GEd technologies. Examples include:

- KALRO: involved in GEd R&D, GEd communication, and extension services.
- Kenyatta University (KU): involved in GEd R&D, education, and research training of BSc, MSc, and PhD students through curricula.
- UoN: involved in education and research training of BSc, MSc, and PhD students through curricula.
- Egerton University (EGU): involved in education and research training of BSc, MSc, and PhD students through curricula.
- JKUAT: involved in education and research training of BSc, MSc, and PhD students through curricula.
- Pwani University (PU), involved in the education and research training of BSc, MSc, and PhD students through curricula.
- SEKU: involved in the education and research training of BSc, MSc, and PhD students through curricula.
- IITA: involved in GEd R&D and capacity building (training), including hands-on training for scientists and regulators.
- ILRI: involved in GEd R&D and capacity building (training) of scientists.
- GALVmed: involved in the commercialization of vaccines developed by ILRI through GEd technologies.
- AGTECH: involved in field-testing and commercial registration of GEd products.
- ISAAA: involved in training of stakeholders in GEd communications and public perception.

3.7.3. Training and Professional Development

As mentioned in section 4.1 below, the IITA has established a state-of-the-art Genome Editing Platform in Nairobi to train African scientists in cutting-edge GEd technologies, such as CRISPR-Cas9 and Cas-CLOVER. Uniquely positioned, the IITA platform, in partnership with international and regional bodies like IGI, AOCC, and Africa Plant Breeding Academy at University of California, Davis (AfPBA, UC Davis) (*See Table 4*), conducts the GEd editing pipeline in-house from target gene identification and guide RNA design to cloning, transformation, regeneration, and molecular characterization of edited plants. Since the

inception of the IITA platform, a broad spectrum of Kenya's stakeholders, including scientists, regulators, and science communicators, have been trained through short-term programs to raise awareness, foster understanding, enable informed decision-making on genome editing, and develop next-generation scientific leaders. Also, through its Capacity Building Initiative, IITA has provided support for PhD students, postdoctoral researchers, and early-career MSc scientists. The platform provides rigorous, hands-on training in genome editing research. Currently, five PhD students are undergoing training in genome editing at IITA-Nairobi, while two other PhD and two MSc students have previously completed similar training for their degree programs (See Figure 3). IITA is collaborating with local universities in Kenya, including KU, JKUAT, and EGU, to further strengthen this effort by providing students with academic mentorship and access to advanced laboratory facilities (See Table 4).

Through this integrated capacity-building approach, IITA's Genome Editing Platform is not only generating scientific innovations but also cultivating a new generation of African leaders in agricultural biotechnology. These trainees are equipped to become future experts and trainers, helping to build and sustain national capacities across Kenya. These Kenyan scientists have acquired the critical skills needed to integrate GEd into national research programs and establish local capacities. Each cohort gets training in three focused, two-week sessions annually.

Other organizations/initiatives like the Alliance for Science in Africa, the Open Forum on Agricultural Biotechnology in Africa (OFAB), and the AUDA-NEPAD/ABNE (GEd Initiative) have been training key stakeholders in Kenya, namely, regulators, scientists, communicators (including journalists), and farmers on genome editing communication through workshops. These workshops run for 1 to 4 days, depending on the need and the consortium hosting them.

Another initiative, the Teaching and Research in (Neuro) science for Development (TReND) in Africa, in partnership with the Kenya Medical Research Institute (KEMRI), Wellcome Trust and NeuroDev Kenya Study, and the Aga Khan University Hospital (AKU), is supporting capacity building in neuroscience education and research by providing equipment, training, and volunteering schemes.

Table 4: Overview of training programmes on genome editing

Institution / Organization	Training Programme	Target Audience / # of Trainees per year	Frequency	Duration	Gaps Identified
International Institute of Tropical Agriculture (IITA)	CRISPR Course	Postdoctoral, Mid-level research scientist	Annually	4 weeks	Need for more hands-on training; funding gaps; limited number of advanced laboratories with state-of-the-art equipment.
	Capacity Building Initiative		Annually	3 months	Need for more hands-on training and more funding
Innovative Genomic Institute (IGI)	CRISPR Course (Joint with IITA)	Postdoctoral, Mid-level research scientist	Annually	4 weeks	Same as in IITA
Africa Plant Breeding Academy at University of California, Davis (AfPBA, UC Davis), and African Orphan Crops Consortium (AOCC)	CRISPR Course (Joint with IITA)	Postdoctoral, Mid-level research scientist	Annually	4 weeks	Same as in IITA
Alliance for Science in Africa	Hands-on in GEd & GEd communication	Farmers, scientists, and communicators	As need arises	6 weeks; 4 days	Public trust and engagement
Open Forum on Agricultural Biotechnology in Africa (OFAB)	Genome editing communication	Regulators, scientists, media, and farmers	Annually	1 to 4 days	Public trust and engagement
AUDA-NEPAD/ABNE (GEEd initiative)	Genome editing communication	Regulators, scientists, and communicators	As need arises	2 to 4 days	Public trust and engagement
Teaching and Research in (Neuro) science for Development (TReND) in Africa	Capacity building in neuroscience	Healthcare scientists or researchers	Annually	8 weeks to 3 years	Curriculum and training, funding, and research infrastructure gaps

In addition, several Kenyan higher institutions of learning, particularly universities, are offering training courses (Certificates, Diplomas, BSc, MSc, and PhD degrees) in biotechnology and its related disciplines, which can be major drivers and leverage for GEd technologies (See Table 5).

Table 5: Universities offering training in biotechnology and GEd-related courses in Kenya

Universities	Certificate and Degree Courses Offered			
	Certificate	BSc	MSc	PhD
Kenyatta University	----	Biotechnology Biochemistry Molecular & Cellular Biology	Biotechnology Biochemistry	Biotechnology Medical Biochemistry
University of Nairobi	----	Food Science and Technology Microbiology and Biotechnology	Plant Breeding & Biotechnology Biotechnology Bioinformatics	Plant Breeding and Biotechnology Biotechnology Bioinformatics
Jomo Kenyatta University of Agriculture and Technology	Basic Microbiological Techniques Molecular Biology	Biotechnology Biochemistry and Molecular Biology Genomic Sciences Industrial Biotechnology	Molecular Biology and Bioinformatics Medical Biotechnology	Molecular Biology and Bioinformatics Molecular Biology and Biotechnology
Pwani University	----	Biotechnology	Biotechnology Bioinformatics	Biotechnology
Egerton University	----	Biochemistry and Molecular Biology Biomedical Science and Technology	Plant Biotechnology Animal Breeding and Genomics	Plant Biotechnology Animal Breeding and Genomics:
University of Embu	----	Microbiology and Biotechnology	Genetics Plant Breeding and Biotechnology Biotechnology Bioinformatics	Biotechnology Bioinformatics Animal Genetics and Breeding
University of Eldoret	----	Biotechnology and Biosafety Agricultural Biotechnology	Plant Breeding and Biotechnology Seed Science and Technology	Plant Breeding Seed Science
Jaramogi Oginga Odinga University	----	Agricultural Biotechnology	Agricultural Biotechnology	Agricultural Biotechnology

of Science and Technology				
Maseno University	----	Medical Biotechnology with IT	Molecular Microbiology & Biotechnology Genetics	Molecular Microbiology & Biotechnology
Masinde Muliro University	----	Biotechnology Molecular Biology and Biotechnology Agricultural Biotechnology	Medical Biotechnology Molecular Biology Genetics and Plant Breeding Bioinformatics	Molecular Biology Bioinformatics

---- No Certificate Courses related to Biotechnology and GEd offered

3.7.3.1. Role of Commission of University Education

The recognition and quality of the certificate and degrees in biotechnology and GEd related courses offered at the Universities listed in Table 5 benefits from the roles of the Commission for University Education (CUE) formerly the Commission for Higher Education (CHE). CUE is the statutory regulatory body in Kenya established under the Universities Act, No. 42 of 2012. The CUE's primary role is to regulate, accredit, and quality-assure university education, ensuring standards are met through accreditation of programs and institutions, policy formulation, and advising the government on education matters. Its core functions are:

- **Accreditation and Licensing:** CUE is the sole authority for accrediting both public and private universities in Kenya, granting Charters or Letters of Interim Authority.
- **Quality Assurance & Inspection:** The commission regularly inspects, monitors, and evaluates universities to ensure compliance with standards and accredits academic programs.
- **Policy Advisory:** It advises the Cabinet Secretary on all matters related to university education.
- **Recognition of Qualifications:** CUE equates and recognizes degrees, diplomas, and certificates from foreign universities to determine their validity in Kenya.
- **Policy & Standards Development:** It develops policies for university admissions, quality assurance, and research standards.
- **Promotion of Research & Innovation:** It encourages and oversees research and innovation to align with national development goals (e.g., Vision 2030).

- **Public Advisory:** CUE informs the public about accredited universities and acts against unaccredited, fraudulent institutions.
- **Licensing Foreign Institutions:** CUE licenses student recruitment agencies and regulates foreign university branches operating in Kenya

3.8. Infrastructure and Equipment

3.8.1. Universities

Most Kenyan universities possess basic molecular biology laboratories and greenhouse facilities. However, these institutions face constraints in laboratory consumables, sequencing services, and procurement processes, which limit their ability to fully support genome-editing research (See Table 6).

Table 6: Status and needs assessment of biosafety laboratory facilities by universities

Institution	Type of Facility	Biosafety Level	Status (Annex A)	Limitations	Support Needed
Kenyatta University	Molecular lab with PCR machines, glasshouses	BSL 2	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable supply of power, and maintenance challenge	Specialized procurement/ waiver/ exemptions
University of Nairobi (CEBIB)	Molecular lab, tissue culture, Bioinformatics	BSL 2	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Egerton University	Molecular lab, tissue culture,	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Pwani University	Molecular lab, tissue culture, glasshouses, Bioinformatics	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Jomo Kenyatta University of	Molecular lab, tissue culture, glasshouses	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding,	Specialized procurement/

Agriculture and Technology				unstable power supply, and maintenance challenges	waiver/ exemptions
University of Embu	Molecular lab, tissue culture, glasshouses	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
University of Eldoret	Molecular lab, tissue culture, glasshouses	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Jaramogi Oginga Odinga University of Science and Technology	Molecular lab, tissue culture, glasshouse	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions
Southeastern Kenya University	Molecular lab, tissue culture, glasshouse	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable power supply, and maintenance challenges	Specialized procurement/ waiver/ exemptions

3.8.2. Research Institutions

National agricultural research organizations and CGIAR centres in Kenya have well-equipped molecular laboratories. Nonetheless, funding limitations and bureaucratic procurement systems often restrict access to specialized equipment, such as high-throughput DNA sequencers. (See Table 7).

Table 7: Status and needs assessment of biosafety laboratory facilities by research institutions

Institution	Type of Facility	Biosafety Level	Status (see Annex A)	Limitations	Support Needed
International Institute of Tropical Agriculture (IITA)	Molecular lab with PCR machines, glasshouses	BSL 2	Fully equipped	Funding gap due to USAID exit	Specialized procurement/ waiver/ exemptions,
International Livestock	Molecular lab with PCR	BSL 2	Fully equipped	Funding gap due to USAID exit	Specialized procurement/

Research Institute (ILRI)	machines, glasshouses				waiver/ exemptions
International Centre of Insect Physiology and Ecology (ICIPE)	Molecular lab with PCR machines, glasshouses	BSL 1	Fully equipped	Funding gap due to USAID exit	Specialized procurement/ waiver/ exemptions
International Maize and Wheat Improvement Center (CIMMYT)	Molecular lab with PCR machines, glasshouses	None	Fully equipped	Funding gap due to USAID exit	Specialized procurement/ waiver/ exemptions
Kenya Agricultural & Livestock Research Organization (KALRO)	Molecular lab with PCR machines, glasshouses	BSL 1, BSL 2	Not fully equipped	inadequate funding,	Specialized procurement/ waiver/ exemptions
Kenya Forestry Research Institute (KEFRI)	Molecular lab with PCR machines, glasshouses	None	Not fully equipped	Political issue (non-enabling national procurement law), inadequate funding, unstable supply of power, maintenance challenge,	Specialized procurement/ waiver/ exemptions

3.8.2.1. Role of National Commission of Science, Technology and Innovation (NACOSTI)

The NACOSTI is the primary state corporation that regulates, coordinates, and promotes research in Kenya under the *Science, Technology, and Innovation (STI) Act No. 28 of 2013*. NACOSTI emphasizes Science, Technology, Engineering, and Mathematics (STEM) development; research into food security and biotechnology, including GEd; and the integration of indigenous knowledge. It also mandates that research involving human subjects or animal testing has approval from accredited Ethics Review Committees. NACOSTI's key functions that potentially benefit the application of modern biotechnology and GEd innovations are:

- **Research Licensing and Authorization:** NACOSTI is responsible for approving and issuing licenses for all scientific research conducted in Kenya, including vetting research proposals.
- **Registration and Accreditation:** The commission registers and accredits research institutions to ensure they maintain high standards and comply with national regulations.

- **Regulation and Quality Assurance:** NACOSTI sets, develops, and enforces codes, guidelines, and regulations governing the management and quality assurance of research, science, technology, and innovation programs.
- **Policy Advisory:** It advises the Kenyan government on policies and matters related to science, technology, and innovation systems.
- **Setting Research Priorities:** In consultation with stakeholders, NACOSTI develops national research priorities in line with the country's economic and social policies.
- **Coordination and Monitoring:** It monitors and evaluates research and technology development activities to ensure efficiency and relevance.
- **Promoting Collaboration:** NACOSTI fosters linkages between academia, industry, the community, and government to enhance technology transfer.

3.9. Traditional and Staple Crops, Livestock, Agroforestry, and Fisheries Varieties/Breeds for Improvement Using GE

The National Agriculture Production Strategy (2023 - 2027) focuses on enhancing food security, accelerating industrialization, and enhancing export growth through key value chains. Several high-impact traditional and staple crops, aimed at reducing import reliance/import substitution, national food security, and climate resilience, are prioritized in the Strategy. During the Assignment, these crops were identified as potential candidates for GE interventions, based on trait targets and socio-economic relevance, including productivity, post-harvest loss reduction, climate adaptation, nutrition, and dietary diversification (See Table 8).

Table 8: National priority crops that could potentially benefit from GEd technology

Crops	Trait improvement of Interest	Socioeconomic Justification	GEd Potential (Low / Medium / High)	Existing R&D	Actual Annual Production (tonnes)	Expected Annual Production (tonnes)
Maize	Yield/ pest/disease/ herbicide tolerance	Staple crop; Central to food and nutrition security	High	CIMMYT–KALRO MLN and striga resistance	4,407,766	5,068, 931
Sorghum	Improved yield and high nutrition	Drought-resilient staple, key in ASAL regions; supports food diversification policy.	High	KU/ICRISAT/partners CRISPR on LGS1 (Striga resistance); national breeding programs	201, 524	241, 829
Millet (Finger & Pearl)	Improved yield and high nutrition	Reduce reliance on maize; policy-supported crops in semi-arid zones	High	ICRISAT CRISPR-enabling work in small millets; trait discovery for stress tolerance	96, 090	115,308
Common bean	Pest and Disease Resistance	A major source of protein and income	Medium	Alliance (CIAT) CRISPR for BCMV/BCMN & bruchid; Kenyan breeding programs for resilience	1, 000, 000	1, 150, 000
Pigeon pea	Pest and Disease Resistance	Important semi-arid legume	Medium	ICRISAT CRISPR platform; trait targets for sterility mosaic & Fusarium—proof-of-concept stage	92, 061	110, 473
Groundnut	Aflatoxin resistance	A major source of protein and income	Medium			
Garden pea	Powdery mildew resistance	A major source of protein and income	Medium			
Mung bean	Pest and Disease Resistance	Drought-adapted pulse; supports dietary diversification	Medium	Emerging CRISPR targets for bruchid resistance	168, 404	202, 085

Banana	Disease resistance	Staple in highland and cash crop zones	Medium	IITA & partners CRISPR for BXW/BBTV	2, 908, 221	3,489, 865
Cassava	Yield	Critical root crop for food security	High	IITA–KALRO genome editing CBSD/CMD resistance	1,202,389	1,502,986
Sweet potato	Pest and Disease Resistance	Nutrient-rich food and cash crop	High	CIP & national partners exploring CRISPR for virus resistance	669,149	802,979
Irish Potato	Pest and Disease Resistance	High-value crop and source of income	High	KALRO–CIP biotech late blight-resistant	2,300,000	2,875,000

Source: National Agriculture Production Strategy (2023 - 2027); National Agriculture Production Report 2024 in the Kenya National Bureau of Statistics

3.10. Intellectual Property Rights and Benefit Sharing

Kenya has established national institutions responsible for managing intellectual property rights (IPRs) related to biotechnology and GEd, addressing issues such as patents, trademarks, and plant breeders' rights through requisite acts/laws as described below:

Kenya Intellectual Property Institute (KIPI): KIPI, through the *Industrial Property Act Cap 509, 2001* (revised 2002, 2007, 2017, 2018, 2022), the Industrial Property Regulations Guidelines for the Examination of Patents, Utility Models and Industrial Designs and the Industrial Property Rights Guidelines for Commercialization in Kenya, promotes innovation and facilitates the acquisition of technology. They provide a legal framework for the protection and regulation of industrial property rights, including patents (e.g., modern biotech/GEd), utility models, and industrial designs. **Kenya Plant Health Inspectorate Service (KEPHIS):** KEPHIS, through the Plant Breeders' Rights and Regulations 2022, grants rights to plant breeders, upholding variety protection, including for varieties developed through modern biotechnology and GEd technologies; sets standards to ensure that quality seeds reach farmers; and prevents the introduction of pests and diseases. **Kenya Copyright Board (KECOBO):** KECOBO is a state corporation and, under the *Copyright Act 2001, CAP 130* (revised 2009), and the Copyright Regulations 2020, is responsible for the administration and enforcement, including registration, infringement, and public awareness of copyright matters. The Copyright Tribunal hears appeals from KECOBO decisions. **Anti-Counterfeit Authority (ACA):** Under the *Anti-Counterfeit Act No. 13, 2008*, the ACA enforces against counterfeiting and related IPR issues. The approaches mentioned above are in line with national, regional, and international agreements/treaties. The main gaps identified for full implementation are funding/investments, political (unexpected political pronouncements), and legal (court cases) issues. Kenya has not handled any IPR-related issues relevant to GEd because GEd products are not yet on the market. Therefore, personnel at national institutions dealing with IPR issues have not been trained on IPR related to GEd. A summary of these IPRs is presented in Table 9.

Note: Although Kenya is a signatory to the Nagoya Protocol and the Nagoya–Kuala Lumpur Supplementary Protocol since 2025, genome editing projects identified in this assessment have not yet been directly linked to these protocols.

Table 9: A summary of IPRs, their institutions and laws in Kenya

Intellectual Property Rights (IPRs)	Organization / Institution	Laws (Acts) / Regulations / Protocols / Treaties
Patents, Utility Models and Industrial Designs; Industrial Property Rights for Commercialization	Kenya Intellectual Property Institute (KIPI)	Industrial Property Act Cap 509, 2001 (revised 2002, 2007, 2017, 2018, 2022)
Plant Breeders Rights	Kenya Plant Health Inspectorate Service (KEPHIS)	Seeds and Plant Varieties Act (CAP 326) 1972 (amended 1991, 2012, 2016, 2025); Plant Breeders' Rights and Regulations, 2022
Copyrights (includes: registration, infringement & public awareness)	Kenya Copyright Board (KECOBO)	Copyright Act 2001 (CAP 130, revised 2009) and Copyright Regulations 2020
Counterfeiting and its IPR-related issues.	Anti-Counterfeit Authority (ACA)	Anti-Counterfeit Act No. 13, 2008

3.11. Private Sector Participation

Private companies play crucial roles in biotech and GEd in Kenya, focusing on commercializing these technologies, often through public-private partnerships (PPPs) to improve both crops and livestock. Private sector roles include funding, developing, and licensing biotech and GEd products (R&D), offering related testing and analytical services, and enabling efficiencies in technology scale-up in agriculture, the environment, and animal health. The companies can also promote these technologies by developing niche biotechnologies, facilitating the transfer of existing technologies, and contributing to the infrastructure needed for biotech/GEEd R&D. A list of key private sector companies in agricultural development in Kenya is presented in Table 10.

Table 10: Overview of private sector companies involved in seed production & distribution, crop improvement & Biotech/GEEd activities in Kenya

Company	Type (Multinational / Regional / National)	Biotech/GEEd Activities	Partnerships	Challenges Faced	Investment Interest
Corteva	Multinational	CRISPR/Cas9 editing of maize on MLN and striga resistance.	KALRO, KU CIMMYT, STAK	Regulatory and legal hurdles (court cases on Seeds and Plant Variety Act)	Crop breeding (Biotech/GEEd & conventional)
Pivot Bio Inc.	Start-up	Suicide plasmid editing of a bacterial strain for self-fertilizing maize	AGTECH, KALRO, Egerton University	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Self-fertilizing cereals (e.g., maize)
Bayer	Multinational	Biotech/No GEEd in Kenya	KALRO, CIMMYT, AATF, etc.	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (Biotech & conventional)
Syngenta	Multinational	No Biotech/GEEd in Kenya	KALRO, CIMMYT	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
SeedCo	Multinational	No Biotech/GEEd	KALRO, CIMMYT	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
Pannar Seed Co. (a Corteva brand)	Multinational	No Biotech/GEEd	KALRO, CIMMYT	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
Kenya Seed	National & Regional	No Biotech/GEEd	KALRO	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
East African Seed Co. (EASEED)	National & Regional	No Biotech/GEEd	KALRO	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
Western Seed Co.	National	No Biotech/GEEd	KALRO, AGRA	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Crop breeding (conventional)
Gicheha Farm Ltd	National	No Biotech/GEEd	KALRO, AATF, AGRA	Regulatory and legal hurdles (e.g., court cases on Seeds and Plant Variety Act)	Production and distribution of maize seeds

Simlaw Seeds	National & Regional	No Biotech/GE	Info not available	Counterfeits, high cost of production, and regulatory hurdles	Production and distribution of vegetable seeds
Amiran Kenya	National	No Biotech/GE	Info not available	Counterfeits, high cost of production, and regulatory hurdles	Production and distribution of vegetable seeds
Kenya Highlands Seed (Royal Seed)	National	No Biotech/GE	Info not available	Counterfeits, high cost of production, and regulatory hurdles	Production and distribution of quality seeds

Private sector involvement in Kenya can be summarized as follows:

- **Collaboration with Public Sector in R&D:** Private sector companies, like Corteva and Bayer, have been actively involved in biotech and GEEd R&D in Kenya. Often, these companies collaborate with public institutions, as will be seen later in this section.
- **Product Development & Innovation:** Private sector in Kenya, as elsewhere, is actively involved in Product Development & Innovation of biotech and GEEd, particularly in agriculture, at the tail end of R&D.
- **Product Testing and Analytics:** In Kenya, the private sector entities contribute to the biotech and GEEd ecosystem by providing specialized services in their advanced laboratories, diagnostic tools, and specialized equipment to the public institutions (partners) for research and commercialization purposes. The growth of private-sector investment and expertise is crucial to enhancing Kenya's capacity to harness biotech and GEEd technologies for agricultural benefits.
- **Commercialization and Scale-up:** Private sector companies are crucial in bringing new biotech/GEEd products to the market and scaling up production/adoption of such products. In Kenya, such products include seeds of improved varieties/hybrids of cultivated crops and animal vaccines.
- **Infrastructure and Service Provision:** The Maize Double-Haploid (DH) facility at Kiboko was established as a joint venture between KALRO and CIMMYT. At the same time, the Maize Lethal Necrosis (MLN) Screening facility at Naivasha was also a KALRO-CIMMYT collaboration, supported by the Bill & Melinda Gates Foundation and the Syngenta Foundation for Sustainable Agriculture. Private sector companies like Monsanto (now Bayer) contributed technical services and infrastructure support, but KALRO-CIMMYT partnerships drove the core establishment. It was established in 2013 to screen maize germplasm for resistance and tolerance to the devastating MLN.
- **Niche Technologies:** In Kenya, private sector entities play an increasingly important role in niche biotech and GEEd technologies through investment in R&D, development of innovative products, and participation in the PPPs as described in detail above. These private sector entities contribute to the ecosystem by addressing specific agricultural needs, such as crop improvement for biotic and abiotic stresses and nutritional quality attributes, thereby supporting the growth of local biotech hubs and positively influencing the development of a robust and friendlier regulatory regime for the very technologies envisaged.
- **Public Private Partnerships (PPPs):** PPPs are a common model for collaborations between public and private institutions. The PPPs bridge gaps in funding and technical expertise, allowing public research to be scaled and commercialized by the private

sector. This enables the private sector to leverage public research capabilities and share any risks in developing appropriate technologies for smallholder farmers (SHFs). Examples of PPP models in Kenya are:

- **Corteva, KALRO, CIMMYT, and STAK Partnership:** CRISPR/Cas9 genome editing of maize for resistance to maize lethal necrosis (MLN) disease in Kenya
- **Pivot Bio, Inc., AGTECH, KALRO, and Egerton University Partnership:** Suicide plasmid genome editing of certain soil bacterial strains that colonize maize root biosphere for nitrogen (N₂) uptake from the atmosphere for self-fertilization. This GEd product is also found to work in other cereals like sugar cane, wheat, etc., of the grass (*Graminae*) family
- **Technology Transfer & Licensing:** To facilitate the development and commercialization of biotech and GEd products, private sector companies in Kenya license (usually non-inclusive) specific technologies and transfer of knowledge to the public research organizations/institutions (e.g., KALRO) and universities (e.g., Kenyatta University).
- **Strengthening Regulatory Capacity and Technical Expertise:** The first early GEd applications to the NBA for approval were partnerships between private sector companies, KALRO, and the international research bodies. These applications helped build regulators' capacity to review GEd products. They accelerated the regulatory approval process, shifting the focus toward a product- and science-based framework rather than a precautionary stance.

Despite the important roles the seed companies play in Kenya, they operate in a highly dynamic, regulated, and competitive environment. While the market has over 200 registered seed companies in Kenya, substantial growth is driven by the demand for improved, climate-resilient, and high-yielding varieties. As a result, the sector is experiencing significant challenges, including a landmark ruling on farmer rights, increased government efforts to curb counterfeit seeds, and a push toward digitization certification. Alongside these challenges, there are also opportunities. The challenges and opportunities for private sector seed companies in Kenya are presented in Table 11.

Table 11: Challenges faced by and opportunities available for private sectors in Kenya

Challenges	Opportunities
Counterfeit and Substandard Seeds: The proliferation of fake, low-quality, or expired seeds remains a significant threat to private producers' reputations and farmer productivity, leading to high economic losses.	Climate-Smart Variety Demand: There is a high, growing demand for early-maturity and drought-tolerant seeds (maize, sorghum, and legumes) suitable for unpredictable rain-fed conditions.
Legal and Regulatory Disruptions: A late 2025/early 2026 court ruling has struck down sections of the Seed and Plant Varieties Act that prohibited farmers from sharing or exchanging non-certified farm-saved seeds. This has created uncertainty for formal seed producers who advocate for strict certification to protect intellectual property and quality.	Digitization of Services: KEPHIS is implementing digital certification and tracing tools to enhance efficiency and reduce costs, allowing companies to improve supply chain transparency and combat fake seeds.
Competition with Government Entities: Private companies face competition from state-linked entities (such as Kenya Seed Company) that benefit from subsidized production, reducing the market share of smaller private firms, especially for maize.	Horticulture and Forage Market Expansion: Beyond maize, there is high potential for investment in high-value horticulture crops (e.g., Hass avocado) and improved forage seeds for the dairy sector.
High Cost of Operations: High certification fees, long variety release timelines, and cross-county cess taxes increase the cost of doing business.	Regional Export Potential: Private companies can capitalize on the growing demand for improved seeds in the Great Lakes region and broader Eastern Africa, particularly through specialized seed production and marketing.
Climate Change & Logistics: Unpredictable weather affects multiplication and storage. Poor road infrastructure in rural areas makes distribution difficult and increases costs.	Public-Private Partnerships (PPPs): Opportunities exist for private firms to partner with research institutions (such as KALRO) to commercialize new, licensed varieties.

3.12. Funding and Investment Landscape

Kenya's genome editing research landscape is externally driven and donor-dependent, with major inputs from the Gates Foundation, USAID and CGIAR/ILRI-linked programs, complemented by industry investments (Corteva, Pivot Bio). To ensure sustainability, increased national funding through NRF and government programs will be critical, alongside

scaling PPP models that link global technology developers with Kenyan research institutions and key stakeholders.

To date, NRF has not allocated dedicated funding to genome editing (GE) research. This reflects limited prioritization of advanced biotechnologies in the national budget, with most support directed toward broader agricultural research and innovation funds.

Former USAID funded the Striga Smart Sorghum for Africa project, which employs genome editing to enhance sorghum resistance to Striga and herbicides. Kenyatta University leads this project in collaboration with partners, including Addis Ababa University.

Gates Foundation is a major financier of GE in Kenya and Africa, with an estimated USD 15M+ invested (2018–2025) in maize genome editing for MLN resistance. Implemented through CIMMYT, KALRO, Corteva, and African universities, this is one of the largest GE projects in Africa. Another project in Kenya is the introduction of broad-spectrum bacterial blight resistance in rice, led by Heinrich Heine University Düsseldorf (Germany). The project is also funded by the Gates Foundation at USD 3M (Phase II). The other partners are KALRO-Mwea, University of Florida (USA), University of Missouri (USA), International Center for Tropical Agriculture (CIAT), Indian Council of Agricultural Research (ICAR), and Nagoya University (Japan). CGIAR Trust Fund has supported reproductive biotechnologies, including genome editing, to develop transgenic goat models for disease control. ILRI (Nairobi) is a key recipient.

The Belgian Development Cooperation has funded a project (2017–2022) on plant stress tolerance using genome editing tools in maize and Arabidopsis, in collaboration with the University of Nairobi. AUDA-NEPAD, while not a direct financier of R&D, provides technical support, policy advocacy, and regulatory harmonization (2019–present) for GE in Africa. Corteva Agriscience is a core licensor of the CRISPR-Cas9 genome-editing tool and has worked with CIMMYT, KALRO, KU, and other partners since 2018. An overview of national and other potential funding sources for genome editing projects/programs is presented in Table 12.

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Table 12: Overview of national and other potential funding sources for genome editing

Funder/Donor	Organization Type	GE Project	Amount (USD)	Duration	Recipient Institution(s)	Area of Focus
National Research Fund (NRF) Kenya	Government	None	None	None	None	None
United States Agency for International Development (USAID)	Development partner	Striga Smart Sorghum for Africa Project	Not disclosed	2020–present	Kenyatta University, African research partners	Crop genome editing – sorghum
Gates Foundation (formerly, Bill & Melinda Gates Foundation – BMGF)	Development partner	Gene editing to control maize lethal necrosis disease (MLND) in Africa for improved maize productivity and grain harvests	~USD 15M+ (regional program)	2018–2025	CIMMYT, KALRO, Corteva, and African universities	Maize lethal necrosis resistance, food security
CGIAR Trust Fund	Development partner	Application of reproductive biotechnologies to develop a transgenic goat as a model for genetic control of animal diseases	Not disclosed	Ongoing	ILRI (Nairobi), University collaborators	Livestock GEd (disease resistance, vaccines)
International Development Research Center (IDRC - Canada)	Development partner	Accelerating African Swine Fever Virus (ASFV) vaccine development via CRISPR-Cas9 and synthetic biology technologies	Not disclosed	2021–2024	ILRI, Canadian labs, African partners	Livestock vaccine R&D

Belgium Development Corporation	Development partner	Modulation of energy homeostasis in maize and Arabidopsis to develop lines tolerant to drought, genotoxic, and oxidative stresses.	Not disclosed	2017–2022	University of Nairobi, Belgian Universities	Plant stress tolerance via GEd
AUDA-NEPAD	Development partner	Communications, advocacy, and regulatory support	Not financial R&D grants; technical/regulatory support	2019–present	African countries, including Kenya, ISAAA	Policy harmonization, capacity building
Pivot Bio	Private sector/Industry	PROVEN [®] 40 Microbial Field Efficacy Trials in Kenya	Not disclosed (corporate R&D)	2021–present	AGTECH Consulting Ltd, KALRO	Edited microbes for maize self-fertilization
Corteva	Private sector/Industry	CRISPR/Cas9 genome editing in maize for MLND resistance	Corporate investment (undisclosed)	2018–present	CIMMYT, KALRO, AfSTA	Maize

4. RECOMMENDATIONS AND POLICY OPTIONS

4.1. Recommendations

Regulatory and policy frameworks

Kenya has created a flexible regulatory framework for genome editing, supported by GEd Guidelines that allow for case-by-case review. This setup promotes testing, adoption, and scaling of genome editing tech in agriculture. Overall, Kenya shows readiness to integrate genome editing into agricultural R&D, product creation, deployment, and commercialization. However, gaps persist, especially in establishing clear pathways for commercialization, including licensing and stewardship of approved genome-edited products. Data from the NBA indicate that some GEd projects in Kenya still involve transgenes and are classified as GMOs. Consequently, researchers involved often delay applying for NBA approval for SDN-1 projects that are progressing to field tests, product development, or commercialization unless they opt for the GMO approval route. Therefore, regulatory reforms should include the following:

- Simplification of import procedures for laboratory supplies and consumables; and
- Improved implementation of the 2006 National Biotechnology Policy.

Additional institutional gaps include limited coordination among academia, industry, government, and the public, resulting in fragmented approaches to the development of biotechnology and GEd.

Capacity building in GEd technologies

The Genome Editing Landscape Analysis revealed capacity gaps in genome-editing research and regulation in Kenya, despite ongoing efforts by organizations such as IITA and IGI. These gaps underscore the need for more hands-on training and enhanced laboratory infrastructure. Therefore, capacity building in Kenya should emphasize comprehensive theoretical and practical training for both scientists and regulators. New initiatives should build on and expand existing platforms, such as the IITA–IGI genome-editing training programs. Supporting and incentivizing trained scientists at PhD, MSc, and BSc levels to share knowledge within their institutions can create a multiplier effect, encouraging broader adoption, dissemination, and scaling of genome editing technologies.

Communication and public engagement

The Landscape Analysis identified communication and public engagement as vital, particularly regarding the NBA's review of genome-edited products and their classification as Site-Directed Nucleases (SDNs). Clear and effective communication is crucial for enhancing public understanding, encouraging responsible use, supporting inclusive policy development, and

accelerating the adoption of GEd technologies to improve food security and nutrition. Kenya should develop comprehensive, innovative public education and awareness initiatives to highlight both the benefits and risks of GEd technologies. These efforts should target the general public, media, policymakers, scientists, politicians, opinion leaders, indigenous companies, local communities, and other relevant groups.

Fostering international and regional collaborations

International and regional collaborations are vital for Kenya to build its genome editing capabilities, boost food security, enhance public health, and drive economic growth. Such partnerships enable knowledge exchange, harmonize regulations, mobilize resources, and ensure fair access to genome editing technologies. Working with regional organizations like AUDA-NEPAD has already helped align biosafety regulations, reduce trade barriers, and foster agricultural innovation. Kenya should keep forging strong links with international research and training centres, universities, expert groups, and the private sector, especially through public-private partnerships. These collaborations will facilitate knowledge sharing, technical support, and responsible use of technology.

Investment in genome editing R&D strategies in Kenya

Kenya's genome-editing R&D investment currently emphasizes developing crops resistant to weeds, drought, and diseases, as well as self-fertilizing varieties, including maize, sorghum, and bananas. It also involves developing vaccines for livestock. These activities are bolstered by specific capacity-building programs and the gradual establishment of regulatory frameworks that differentiate genome-edited products from traditional GMOs. Despite limited funding, investments are increasingly channelled into infrastructure and training for CRISPR-Cas9 technology. To strengthen investment in genome editing R&D, Kenya should:

- Prioritize development of advanced laboratory infrastructure and sustained training for scientists and regulators;
- Promote regional and international collaborations;
- Foster public trust through transparent and inclusive communication; and
- Develop flexible, harmonized biosafety frameworks that encourage innovation.

Targeted financial incentives for PPPs, investment in young scientists, and engagement with indigenous crop systems will further support sustainable and locally relevant agricultural innovation.

4.2. Policy Options

Summary matrix of action plans

Issues → Gaps → Recommended Actions → Lead Institutions

- i. **Regulatory frameworks** → Policy implementation gaps → Strengthen implementation of the 2006 National Biotechnology Policy and improve stakeholder coordination → NBA
- ii. **Capacity building in GEd R&D** → Knowledge and skills gaps → Expand hands-on training for regulators and scientists; establish advanced public-sector laboratories comparable to IITA platforms → Public research institutions and universities
- iii. **Communication and public engagement** → Limited understanding of GEd vs GMOs → Simplified, targeted communication strategies and public education → KALRO, AUDA-NEPAD, EAC
- iv. **International and regional collaborations** → Partnership gaps → Incentivize PPPs and cross-border collaboration → National government, private sector, foundations (e.g. Gates Foundation)
- v. **Investment in GEd R&D** → Funding constraints → Invest in infrastructure, training, and collaborative platforms → AUDA-NEPAD, EAC, private foundations

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ANNEXES

Annex 1. List of key stakeholders and their institutions in Kenya

SN	Name	Sector	Ministry/Department/Institution/Organization	Position	Contact Details
1	Mr. Nehemiah Ng'etich	Regulatory	National Biosafety Authority (NBA)	Ag. CEO	nngetich@biosafetykenya.go.ke
2	Mr. Josphat Muchiri	Regulatory	National Biosafety Authority (NBA)	Director Biosafety Research and Compliance	jmuchiri@biosafetykenya.go.ke
3	Ms. Ann Muia	Regulatory	National Biosafety Authority (NBA)	Senior Biosafety Officer	ndanuann@gmail.com
4	Dr. Kisa J Z Juma Ngeiywa	Regulatory	Department of Veterinary Services (DVS)	Director	khalifabbey@gmail.com.
5	Prof. Theophilus M. Mutui	Regulatory	Kenya Plant Health Inspectorate Services (KEPHIS)	Managing Director	director@kephis.org
6	Dr. Alexander Muvea	Regulatory	Kenya Plant Health Inspectorate Services (KEPHIS)	In charge of KSTCIE	kephisinfo@kephis.org
7	Mr. Fredrick N. Muchiri	Regulatory	Pest Control Products Board (PCPB)	CEO	md@pcpb.go.ke
8	Dr. Paul Ngaruyia	Regulatory	Pest Control Products Board (PCPB)	Director Research	info@pcpb.go.ke
9	Mr. Barasa Wanyonyi	Regulatory	Pest Control Products Board (PCPB)	Head of Registration	info@pcpb.go.ke
10	Dr. Mamo Boru Mamo	Regulatory	National Environment Management Authority (NEMA)	Director General	info@nema.go.ke
11	Dr. Martin Mwirigi	Research Institutions	Kenya Agricultural and Livestock	Institute Director, Biotechnology	martin.mwirigi@kalro.org

			Organization (KALRO)	Research Institute - KALRO	
12	Dr. James Karanja	Research Institutions	Kenya Agricultural and Livestock Organization (KALRO)	Senior Research Officer	jakakah79@gmail.com
13	Dr. Leena Tripathi	Research Institutions	International Institute of Tropical Agriculture (IITA)	Director of Eastern Africa Hub and Leader of the Biotechnology Program	l.tripathi@cgiar.org
14	Dr. Jaindra Nath Tripathi	Research Institutions	International Institute of Tropical Agriculture (IITA)	Plant Biotechnologis t	J.Tripathi@cgiar.org
15	Dr. Valentine Otang Ntui	Research Institutions	International Institute of Tropical Agriculture (IITA)	Scientist	ntuival@yahoo.com
16	Dr. Colleta Nabwile Khaemba	Research Institutions	International Maize and Wheat Improvement Center (CIMMYT)	Research associate	nabwilekhaemba@gmail.c om
17	Dr. Yoseph Beyene	Research Institutions	International Maize and Wheat Improvement Center (CIMMYT)	Regional Maize Breeding Coordinator for Africa and Maize Breeder for Eastern Africa	y.beyene@cgiar.org
18	Dr. Pieter Rutsaert	Research Institutions	International Maize and Wheat Improvement Center (CIMMYT)	Seed systems specialist	p.rutsaert@cgiar.org
19	Dr. Pauline Muindi	Research Institutions	International Maize and Wheat Improvement Center (CIMMYT)	Research support specialist	P.muindi@cgiar.org

20	Dr. Rose Mburu	Research Institutions	International Maize and Wheat Improvement Center (CIMMYT)	Research Program Administrator	r.mburu@cgiar.org
21	Dr. Hussein Abkallo	Research Institutions	International Livestock Research Institute (ILRI)	Scientist - Molecular Biology & Head of Biosciences Laboratory	Hussein.Abkallo@ed.ac.uk
22	Dr Rebbie Harawa	Research Institutions	International Crops Research Institute for the Semi-Arid Tropics (ICRISAT)	Agricultural Research and Development Expert	icrisat-ken@icrisat.org
23	Dr. Jane Mbugua	Research Institutions	Kenya Agricultural and Livestock Organization (KALRO)	Chief Research Scientist	jayney480@yahoo.com info@kalro.org director@kalro.org
24	Dr. Kouadio Nasser Yao	Research Institutions	International Livestock Research Institute (ILRI)	Senior Scientist BecA Task Force Lead & DLB Pan-African Coordinator	N.Yao@ilri.org
25	Dr. Emily Waringa Gichuhi	Research Institutions	Kenya Agricultural and Livestock Research Organization (KALRO)	Senior Scientist.	e.gichuhi@kalro.org
26	Dr. Paul Kuria	Research Institutions	Kenya Agricultural and Livestock Research Organization (KALRO)	Senior Scientist.	p.kuria@kalro.org
27	Todd Jones	Private sector/Industries & NGOs	Corteva Agriscience	Research Director	todd.j.jones@corteva.com
28	Margaret Karembu	Private sector/Industries & NGOs	Agri- biotech Applications (ISAAA AfriCenter)	Environmental Science	mkarembu@isaaa.org.

				Management Specialist	
29	Joyce Wairimu Karanja	Other Government Institutions & Agencies	Seed Trade Association of Kenya (STAK)	Secretariat Lead	joycekaranja@stak.or.ke
30	Prof. Steven Runo	Universities	Kenyatta University	Associate Professor in Molecular Biology, Biotechnology	runo.steve@ku.ac.ke
31	Prof. Mathew Piero	Universities	Kenyatta University	Associate Professor	piero.mathew@ku.ac.ke
32	Prof. George Ogello Obiero	Universities	University of Nairobi	Director, Center for Biotechnology and Bioinformatics (CEBIB)	george.obiero@uonbi.ac.ke
33	Dr. Wilkister Nakami Nabulindo	Universities	University of Nairobi	Veterinary surgeon	wilkisterkelly@gmail.com
34	Prof. James Nguhiu-Mwangi	Universities	University of Nairobi	Large Animal Surgery	jnmwangi@uonbi.ac.ke
35	Prof. Wilton Mbinda	Universities	Pwani University	Associate professor	w.mbinda@pu.ac.ke
36	Dr. Cecilia Mweu	Universities	Jomo Kenyatta University of Agriculture and Technology	Director of the Institute of Biotechnology Research (IBR)	cmweu@jkuat.ac.ke
37	Bicko Steve Juma	Universities	Jomo Kenyatta University of Agriculture and Technology	PhD student	bickosnr@agro.au.dk
38	Prof. Paul Njiruh Nthakanio	Universities	University of Embu	Associate Professor	nthakanio.paul@embuni.ac.ke

39	Prof. George Obiero	Universities	University of Nairobi	Director, Center for Biotechnology and Bioinformatics (CEBIB)	george.obiero@uonbi.ac.ke.
40	Prof. E. Nguu	Universities	University of Nairobi	Associate Professor, Biochemistry Department	ednguu@uonbi.ac.ke
41	Prof. Wilkister Kelly	Universities	University of Nairobi	Veterinary surgeon	wilkisterkelly@gmail.com
42	Prof. J.N. Mwangi	Universities	University of Nairobi	Large Animal Surgery	jnmwangi@uonbi.ac.ke
43	Prof. W. Mbinda	Universities	Pwani University	Associate professor	w.mbinda@pu.ac.ke.
44	Prof. Oduor Richard	Universities	Kenyatta University	Professor in Molecular Biology, Biotechnology	oduor.richard@ku.ac.ke
45	Prof. Njeru Ezekiel	Universities	Kenyatta University	Associate Professor in Microbiology	njeru.ezekiel@ku.ac.ke
46	Prof. Ngeranwa	Universities	Kenyatta University	Professor	ngeranwa@gmail.com
47	Prof. Miano	Universities	University of Nairobi	Professor in Molecular Plant Virology	dwatuku@yahoo.com
48	Prof. Kahi Ngugi	Universities	University of Nairobi	Professor in Molecular Biology, Biotechnology	kahiukahiu@gmail.com
49	Prof. George Obiero	Universities	University of Nairobi	Director, Center for Biotechnology and Bioinformatics (CEBIB)	george.obiero@uonbi.ac.ke

50	Prof. S.M. Githiri	Universities	Jomo Kenyatta University of Agriculture and Technology	Professor of Genetics and Plant Breeding	smgithiri@yahoo.com
51	Dr. S. Anami	Universities	Jomo Kenyatta University of Agriculture and Technology	Associate professor	syanakuti@gmail.com
52	Dr. C. Orek	Universities	Muranga University	Molecular plant breeding	corek@mut.ac.ke
53	Dr. Calleb Olweny	Universities	Jaramogi Oginga Odinga University of Science and Technology	Molecular plant breeding	callebolweny2019@gmail.com
54	Prof. Levi Mudogo	Universities	Masinde Muliro University of Science and Technology	Molecular plant breeding	l.mudogo@gmail.com
55	Dr. E. Kuria	Universities	Chuka University	Molecular plant breeding	ekuria@chuka.ac.ke
56	Dr. Rose Ramkat	Universities	Moi University	Molecular Biological Sciences	rose.ramkat@mu.ac.ke
57	Prof. J. Ogwen	Universities	Egerton University	Associate Professor of Horticulture/Biotech	jogweno@egerton.ac.ke
58	Prof. Stephen Indieka	Universities	Egerton University	Molecular Biological Sciences	stephen.indieka@egerton.ac.ke
59	Prof. M. Dida	Universities	Maseno University	Professor of Genetics and Plant Breeding	mitodida@yahoo.com
60	Dr. Okanya	Universities	Technical University of Kenya	Biochemistry Department	okanyap@tukenya.ac.ke
61	Dr Stephen Kironji Githeng'u	Universities	Egerton University	Crop Science	stephen.githengu@egerton.ac.ke
62	Prof. M. Mwamburi	Universities	Taita Taveta University	Molecular plant breeding	mwamburim@gmail.com

63	Dr. Edith Mnyazi Muwawa	Universities	Pwani University	Senior Lecturer/ Researcher	edithmuwawa@yahoo.co m
64	Dr. Grace Wacheke Mungai	Universities	Jomo Kenyatta University of Agriculture and Technology	Senior Lecturer.	gmungai@ibr.jkuat.ac.ke